PORTFOLIO

HARINI P

B.Tech CSE (Big Data Analytics), 2027



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AI and Data Engineering Student

Final-year B.Tech student in Computer Science and Engineering (Big Data Analytics) at SRM Institute of Science and Technology, Vadapalani. My work sits where machine learning meets interface design: classifiers that explain themselves, dashboards that make dense data readable, and tools that turn messy inputs into clear results.

EDUCATION

B.Tech, Computer Science and Engineering (Big Data Analytics)
SRM Institute of Science and Technology, Vadapalani
2023 - 2027 | CGPA 8.26 / 10 (up to Semester 6)

SKILLS

AI and ML	PyTorch, Scikit-learn, XGBoost, MLflow, Grad-CAM, Transfer learning, Explainable AI
Interfaces and web	React, Streamlit, Flask, Django, Power BI
Data	Python, SQL, Pandas, NumPy, MySQL, Git/GitHub

CERTIFICATIONS

Introduction to Machine Learning, NPTEL Elite, IIT Kharagpur (Sep 2025)
Programming in Java, NPTEL Elite (May 2026)
Introduction to Artificial Intelligence, IBM on Coursera (Feb 2025)
Java DSA, Certificate of Excellence, Scalar (Apr 2025)

CONTACT

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GitHub	github.com/harinirenu09-bot

Languages: Tamil (native), English (professional working proficiency)

Experience

SEP 2026 | 1 MONTH

Artificial Intelligence Intern

XTRAGRAD

- Independently designed, trained and deployed a transfer-learning EfficientNet-B0 classifier that detects real vs AI-generated images, reaching 84.70% accuracy on a balanced 2,000-image test set.
- Added Grad-CAM to show the image regions behind each prediction, plus forensic and FFT frequency-domain signals for extra interpretability.
- Built and deployed a public, interactive Streamlit app for real-time classification, and ran a robustness study across JPEG compression, blur, brightness and resize changes.

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JUN 2025 | 4 WEEKS

Volunteer

Ayaan Foundation

Tree planting, beach cleaning and community cleaning drives.

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Explainable classifier and live app

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Fraud detection with model tracking

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Meat safety analytics, React front end

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Rule-based data validation tool

01 AI IMAGE DETECTION

Explainable analysis of real vs AI-generated images | AI internship, XTRAGRAD, Sep 2026

AI-generated images are getting hard to spot, and a bare "fake" label gives people no reason to trust it. I designed, trained and deployed an EfficientNet-B0 classifier that separates real images from AI-generated ones, and shows **why** it decided, using Grad-CAM heatmaps, forensic cues and frequency-domain (FFT) signals. The classifier runs inside a public, interactive Streamlit app for real-time checks.

84.70%
accuracy on a balanced test set

2,000
test images: 1,000 real, 1,000 AI-generated

4
perturbations in the robustness study

Tools: Python, PyTorch, Torchvision, Grad-CAM, Streamlit

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How it works



Explainability

Grad-CAM highlights the regions of an image that drove each prediction, so a verdict comes with visible evidence. Forensic and FFT frequency-domain signals add a second view of the same image for extra interpretability.

Robustness study

To find the model's limits, I tested how accuracy changes when images are altered in everyday ways:

- JPEG compression
- Blur
- Brightness
- Resize

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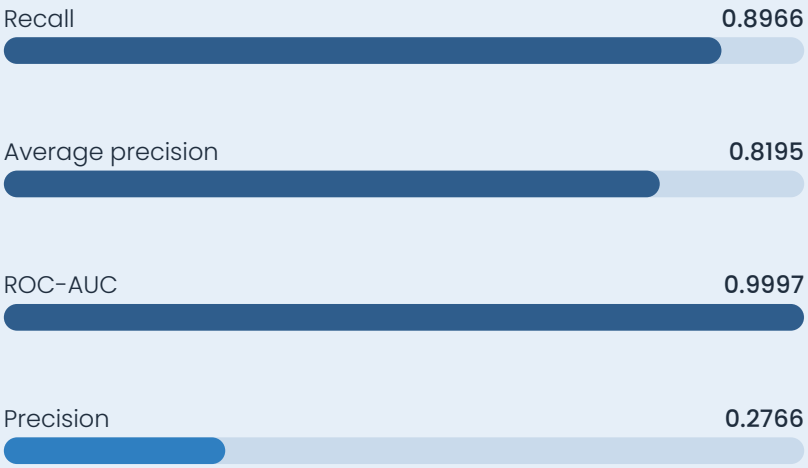
02 FRAUDSHIELD ML

Intelligent financial transaction fraud detection | Team project with Roweena F, SRM

Fraud is rare compared with normal transactions, so manual review and fixed rules are hard to scale. FraudShield ML trains an **XGBoost** model on 200,000 records of the PaySim synthetic dataset, tracks the experiment in **MLflow**, and serves it through a Streamlit app that estimates the fraud probability of a single transaction.

My part: data preparation, XGBoost training, MLflow configuration and review, and model testing. We built the Streamlit app and deployed it together.

Results on the held-out test set



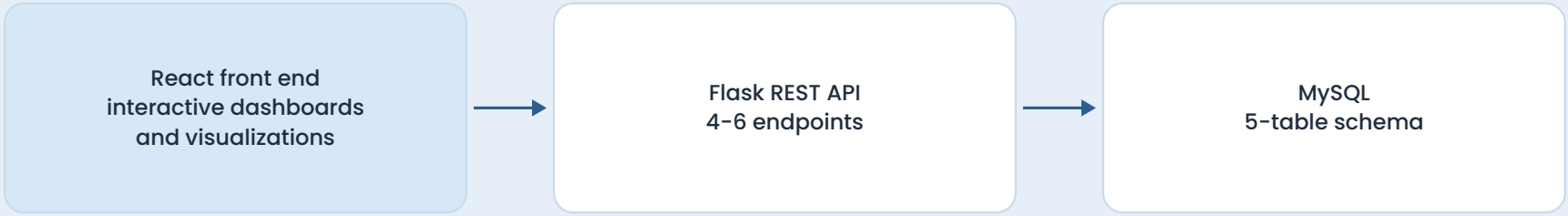
Precision is modest at the 50% threshold, so many alerts are false alarms. Data is synthetic, and monitoring covers one session only.

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03 AGROSAFE DASHBOARD

Smart meat safety and antibiotic monitoring | Team project

Meat safety teams need to track antibiotic usage, withdrawal periods, residue risk and regulatory compliance in one place. Agrosafe is a full-stack analytics dashboard built for that. I built the **React** front end: interactive dashboards and visualizations connected to a Flask REST API, and I worked with teammates on the compliance and safety data model behind the reports.



Highlighted: the part I built.

Tracks

- Antibiotic usage
- Withdrawal periods
- Residue risk
- Regulatory compliance

04 XENO VALIDATOR

Transaction data validation tool | Solo project

Transaction data arrives with bad phone numbers, dates, payment modes and amounts, and errors are slow to find by hand. Xeno Validator checks a file against **15 automated rules** and gives back clean, usable outputs. I tested it on datasets of up to 50,000 rows.



Rules cover

- Phone numbers (country-specific)
- Order dates
- Payment modes
- Quantities
- Amounts

Tools: Python, Streamlit

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Thank you

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